

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

- 1) The James Webb telescope is mainly powered by _____
A) solar energy B) radiation pressure
C) fossil fuel D) RTG (nuclear fission)
- 2) The James Webb telescope is a _____
A) refractive telescope B) a reflective telescope
- 3) The alpha spectral line is a UV line. If the object containing hydrogen moves away from us, the line could be: _____
A) in the red/IR range B) in the X-ray range
C) in the gamma-rays range D) still in the UV range
- 4) Which planet is orbited by the four "Galilean moons" known individually as Io, Europa, Ganymede, and Callisto? _____
A) Neptune B) Uranus C) Saturn D) Jupiter E) Mars
- 5) The James Webb telescope, will orbit a point called _____
A) Galileo point B) Lagrange point C) Kepler point D) Newton point
- 6) The Oort cloud is _____
A) the cloud of gas from which the solar system formed.
B) a region in the same plane as the planets that contains hundreds of thousands of comets.
C) a circular region between the orbits of Mars and Jupiter that contains most known asteroids.
D) a roughly spherical region larger than the orbit of Pluto that may contain a trillion comets.
E) a vast cloud of gas that encircles the solar system at a great distance from the Sun.
- 7) The most massive planet in our solar system is _____
A) Jupiter. B) Neptune. C) Uranus. D) Mars. E) Saturn.
- 8) Which planet has the highest *average* surface temperature and why? _____
A) Venus, because of its dense carbon dioxide atmosphere
B) Mercury, because of its dense carbon dioxide atmosphere
C) Jupiter, because it is so big
D) Mars, because of its red color
E) Mercury, because it is closest to the Sun
- 9) Which of the following indicates that past conditions on Mars may have been suitable for life? _____
A) There are dried up riverbeds, rock-strewn floodplains, and minerals that form in water.
B) Mars has polar caps.
C) There are very large extinct volcanoes on Mars.
D) Mars has two small Moons, Phobos, and Deimos.
E) Mars has been studied by numerous spacecraft, including both orbiters and landers.
- 10) Radio telescopes are generally _____ than visible-light telescopes. _____
A) much smaller B) located at much higher altitudes
C) much more expensive D) much larger

- 11) Suppose two stars are separated in the sky by 0.1 arcseconds. If you take a picture of them with the SOAR telescope, which has an angular resolution of 0.6 arcsecond, what will you see? 11) _____
A) one point of light that is the blurred image of both stars
B) two separate points of light
C) nothing at all
- 12) What is unusual about the planet Venus? 12) _____
A) It orbits the Sun in a direction opposite that of all the other planets.
B) It does not orbit the Sun in the same plane as the other planets.
C) It orbits the Sun faster than it should give the size of its orbit.
D) It rotates extremely rapidly around its axis.
E) It rotates in a direction opposite to the direction of its orbit.
- 13) According to the nebular theory, what are asteroids and comets? 13) _____
A) the shattered remains of collisions between moons
B) chunks of rock or ice that condensed long after the planets had formed
C) leftover planetesimals that never accreted into planets
D) the shattered remains of collisions between planets
E) chunks of rock or ice that have been expelled from the planets by volcanoes
- 14) Which planet is most similar in size and mass to Earth? 14) _____
A) Jupiter B) Neptune C) Mars D) Mercury E) Venus
- 15) What is the purpose of *adaptive optics*? 15) _____
A) to improve the angular resolution of telescopes in space
B) to reduce the distorting effects of atmospheric turbulence for telescopes on the ground
C) to increase the collecting area of telescopes on the ground
D) to increase the magnification of telescopes on the ground
E) to allow several small telescopes to work together like a single larger telescope
- 16) Which of the following would be the most useful ultraviolet telescope? 16) _____
A) a 30-meter telescope on a mountain B) a 2.5-meter telescope in space
C) a 10-meter telescope on a mountain D) a 1-meter telescope in space
- 17) The James webb teslescope will collect light mainly in the ... range 17) _____
A) X-ray B) visible C) UV D) IR
- 18) What causes stars to twinkle? 18) _____
A) Interstellar gas along the line of sight to the star absorbs different amounts of radiation.
B) Stars constantly vary in brightness as they expand and contract.
C) The human eye has difficulty seeing faint objects.
D) Moving gases in Earth's atmosphere randomly bend light rays from stars.
- 19) Comets are made mostly of 19) _____
A) rock and ice.
B) rock and metal.
C) rock.
D) metal.
E) hydrogen gas.

- 20) parallel rays collected by a concave mirror all focus at a single point 20) _____
 A) It depends B) True C) False
- 21) Which statement about Pluto is *not* true? 21) _____
 A) It is the most massive object in the Kuiper belt.
 B) It is made mostly of rock and ice.
 C) Its surface temperature is very cold.
 D) It is both smaller and less massive than Earth.
 E) It was visited by the *New Horizons* spacecraft in 2015.

SHORT ANSWER. Write the word or phrase that best completes each statement or answers the question.

- 22) When a cloud spins, it will spin faster at the center (where there is more mass). This is because of conservation of: 22) _____

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- 23) Earth's atmosphere is most transparent to all wavelengths of _____ light. 23) _____
 A) x-ray
 B) infrared
 C) visible
 D) ultraviolet
 E) both visible and ultraviolet
- 24) If the size of the primary mirror (or lens) of a telescope is multiplied by 3, the brightness is multiplied by 24) _____
 A) 4 B) 6 C) 9 D) 3
- 25) How do the light-collecting area and best possible angular resolution of telescope with a 10-meter diameter mirror compare to that of a telescope with a 5-meter diameter mirror? 25) _____
 A) The 10-meter telescope has double the light collecting area, and its angular resolution is better by a factor of 2.
 B) The 10-meter telescope has 10 times the light collecting area, and its angular resolution is better by a factor of 5.
 C) The 10-meter telescope has 4 times the light collecting area, and its angular resolution is better by a factor of 4.
 D) The 10-meter telescope has 4 times the light collecting area, and its angular resolution is better by a factor of 2.
- 26) Which of the following is farthest from the Sun? 26) _____
 A) a comet in the Kuiper belt
 B) Pluto
 C) a comet in the Oort cloud
 D) an asteroid in the asteroid belt
 E) Neptune
- 27) Sirius is called the 27) _____
 A) dog star B) Bear star C) Orion Star D) Taurus star

- 28) At the equator, Polaris is seen 28) _____
A) at the horizon B) above your head
C) Between the zenith and the horizon D) Can not be seen
- 29) What is true about the distance Earth-Sun 29) _____
A) distance is larger in July B) distance is larger in January
C) It stays constant D) we don't know
- 30) Equinox 30) _____
A) when the day and the night have the same length
B) its the hottest day of the year
C) its the coolest day of the year
D) its the shortest day of the year
- 31) The angle between Polaris and the horizon plane is the 31) _____
A) altitude B) longitude C) right ascension D) tilt of the EArth

Answer Key

Testname: TEST2

- 1) A
- 2) B
- 3) A
- 4) D
- 5) B
- 6) D
- 7) A
- 8) A
- 9) A
- 10) D
- 11) A
- 12) E
- 13) C
- 14) E
- 15) B
- 16) B
- 17) D
- 18) D
- 19) A
- 20) B
- 21) A
- 22)
- 23) C
- 24) C
- 25) D
- 26) C
- 27) A
- 28) A
- 29) A
- 30) A
- 31) A